STA 445 S26 Assignment 5

2026-03-19

Problem 1

Convert the following to date or date/time objects.

a. September 13, 2010.

```
(mdy("September 13, 2010"))
```

```
## [1] "2010-09-13"
```

b. Sept 13, 2010.

```
(mdy("Sept 13, 2010"))
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

This abbreviation for September is incorrect.

c. Sep 13, 2010.

```
(mdy("Sep 13, 2010"))
```

```
## [1] "2010-09-13"
```

d. S 13, 2010. Comment on the month abbreviation needs.

```
(mdy("S 13, 2010"))
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

This abbreviation for September is incorrect.

e. 07-Dec-1941.

```r
(dmy("07-Dec-1941"))
```

```
## [1] "1941-12-07"
```

f. 1-5-1998. Comment on why you might be wrong.

```r
(mdy("1-5-1998"))
```

```
## [1] "1998-01-05"
```

It is not clear whether the date is January 5th or May 1st.

g. 21-5-1998. Comment on why you know you are correct.

```r
(dmy("21-5-1998"))
```

```
## [1] "1998-05-21"
```

This must be correct because there isn't a 21st month in a year.

h. 2020-May-5 10:30 am

```r
(ymd_hm("2020-May-5 10:30 am"))
```

```
## [1] "2020-05-05 10:30:00 UTC"
```

i. 2020-May-5 10:30 am PDT (ex Seattle)

```r
(ymd_hm("2020-May-5 10:30 am PDT", tz="US/Pacific"))
```

```
## [1] "2020-05-05 10:30:00 PDT"
```

j. 2020-May-5 10:30 am AST (ex Puerto Rico)

```r
(ymd_hm("2020-May-5 10:30 am AST", tz="America/Puerto_Rico"))
```

```
## [1] "2020-05-05 10:30:00 AST"
```

Problem 2

Using just your date of birth (ex Sep 7, 1998) and today's date calculate the following:

a. Calculate the date of your 64th birthday.

```r
aa <- mdy("03-17-2007")
(aa + years(64))
```

```
## [1] "2071-03-17"
```

```r
(wday(aa + years(64), label=TRUE, abbr=FALSE))
```

```
## [1] Tuesday
## 7 Levels: Sunday < Monday < Tuesday < Wednesday < Thursday < ... < Saturday
```

b. Calculate your current age (in years).

```r
aa <- mdy("03-17-2007")
bb <- today()
cc <- as.period(aa %--% bb) %>%
  year()
cc
```

```
## [1] 19
```

c. Using your result in part (b), calculate the date of your next birthday.

```r
dd <- aa + years(cc + 1)
wday(dd, label=TRUE, abbr=FALSE)
```

```
## [1] Wednesday
## 7 Levels: Sunday < Monday < Tuesday < Wednesday < Thursday < ... < Saturday
```

d. The number of *days* until your next birthday.

```r
bday <- aa %--% (aa + dyears(1))
ee <- as.period(bday, unit="days") %>%
  day()
ee
```

```
## [1] 365
```

e. The number of *months* and *days* until your next birthday.

```r
bday <- aa %--% (aa + dyears(1))
ee <- as.period(bday, unit="months")
ee
```

```
## [1] "11m 28d 6H 0M 0S"
```

Problem 3

Suppose you have arranged for a phone call to be at 3 pm on May 8, 2015 at Arizona time. However, the recipient will be in Auckland, NZ. What time will it be there?

```
aa <- mdy_hm("May 8, 2015 3:00 pm", tz="US/Arizona") %>%
  with_tz(tz="NZ")
aa
```

```
## [1] "2015-05-09 10:00:00 NZST"
```

Problem 4

It turns out there is some interesting periodicity regarding the number of births on particular days of the year.

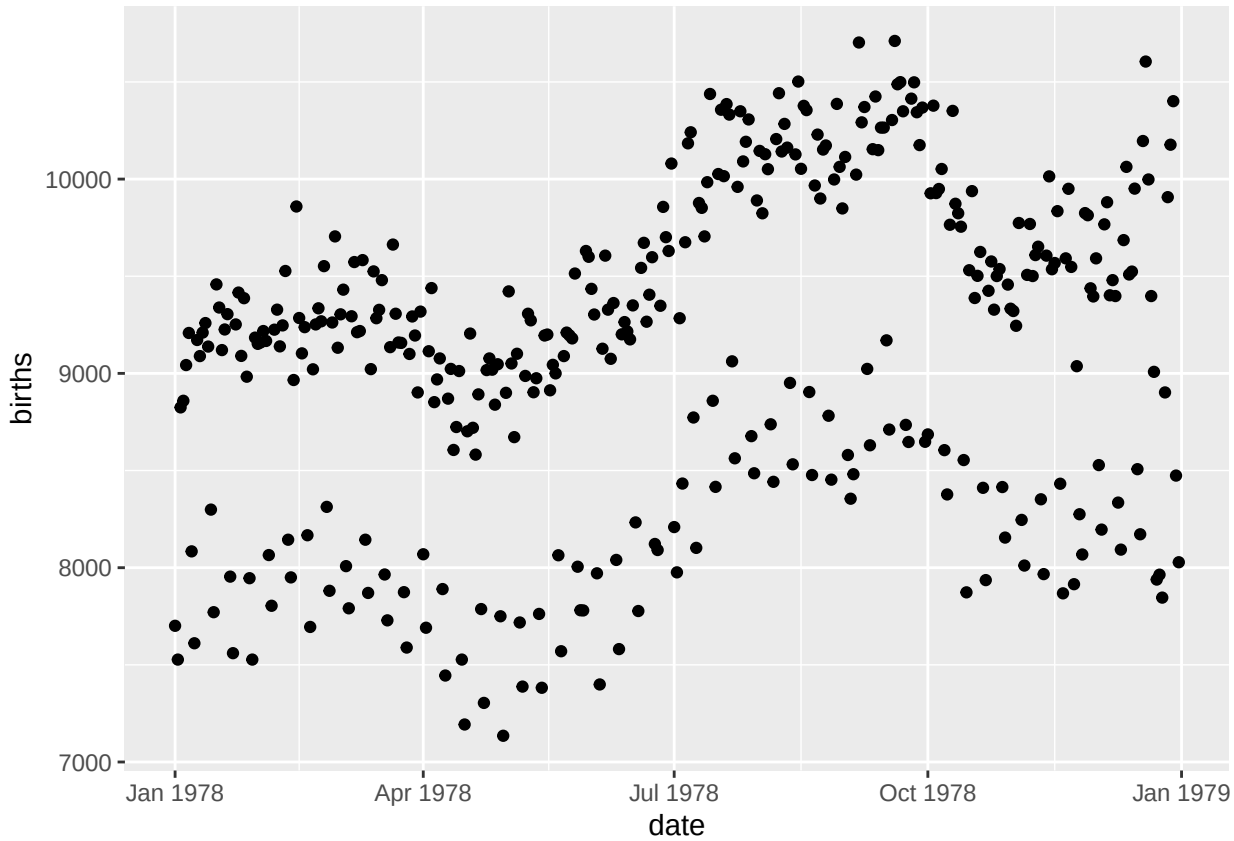
a. Using the `mosaicData` package, load the data set `Births78` which records the number of children born on each day in the United States in 1978. Because this problem is intended to show how to calculate the information using the `date`, remove all the columns *except* `date` and `births`.

```
Births <- Births78
newBirths <- Births[,1:2]
head(newBirths)
```

```
##         date births
## 1 1978-01-01   7701
## 2 1978-01-02   7527
## 3 1978-01-03   8825
## 4 1978-01-04   8859
## 5 1978-01-05   9043
## 6 1978-01-06   9208
```

b. Graph the number of `births` vs the `date` with date on the x-axis. What stands out to you? Why do you think we have this trend?

```
ggplot(newBirths,
       aes(date, births))+
  geom_point()
```



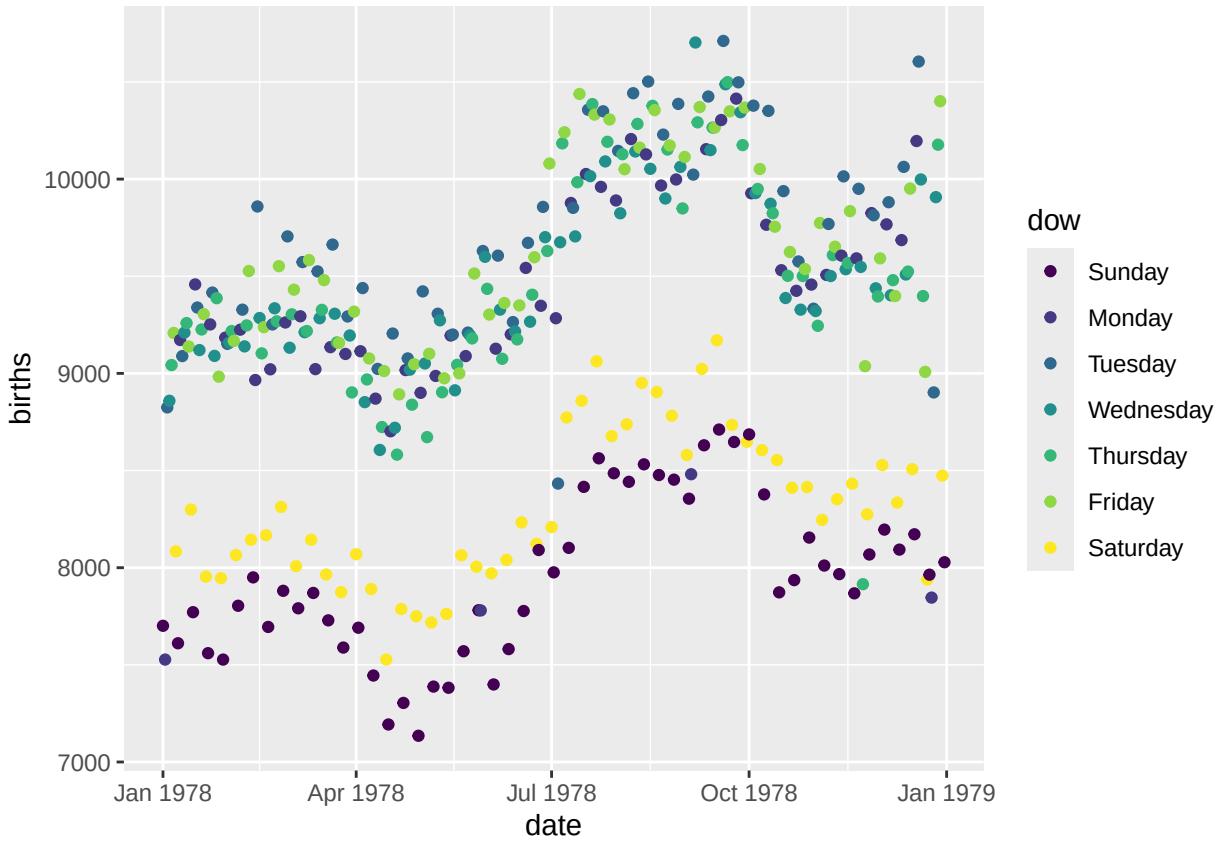
There are two clumps of points that are relatively uniform in change over time. This might be because there are different amounts of births by each weekday but the births on each weekday all change at the same rate over time.

c. To test your assumption, we need to figure out the what day of the week each observation is. Use `dplyr::mutate` to add a new column named `dow` that is the day of the week (Monday, Tuesday, etc). This calculation will involve some function in the `lubridate` package and the `date` column.

```
newBirths <- newBirths %>%
  mutate(dow = wday(ymd(date), label=TRUE, abbr=FALSE))
```

d. Plot the data with the point color being determined by the day of the week variable.

```
ggplot(newBirths,
       aes(date, births, color = dow))+
  geom_point()
```


10000
9000
8000
7000
births
Jan 1978
Apr 1978
Jul 1978
Oct 1978
Jan 1979
date
dow
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday